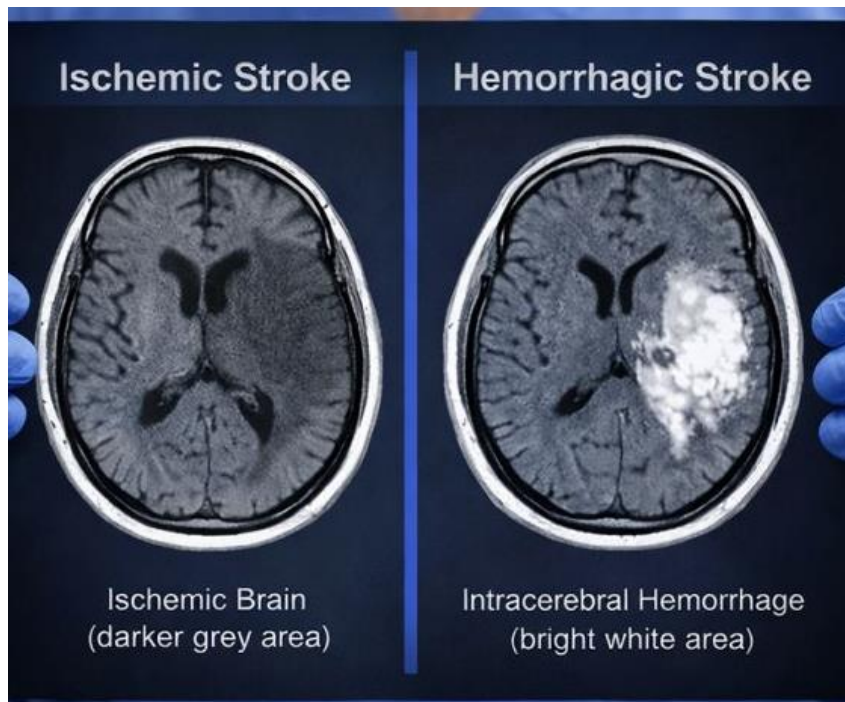


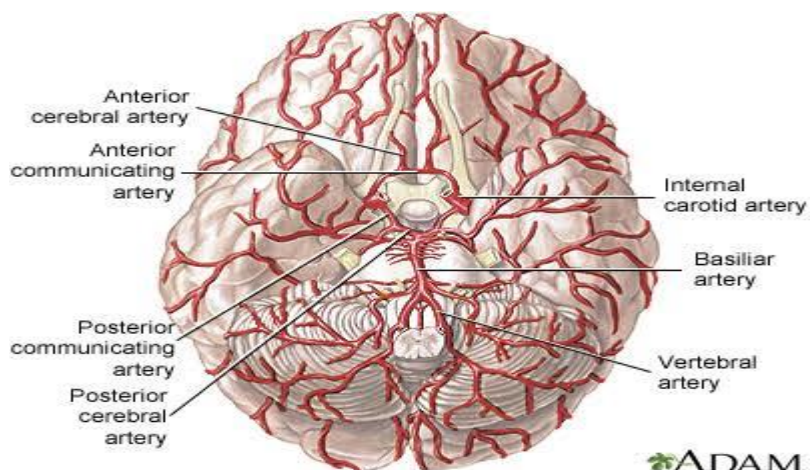
Stroke

Stroke is an acute neurological emergency caused by interruption of cerebral blood flow, leading to brain cell death. It is classified as **ischemic stroke** (blocked artery) or **hemorrhagic stroke** (ruptured blood vessel).



Brain Regions

- Middle cerebral artery territory
- Anterior cerebral artery territory
- Posterior cerebral artery territory
- Brainstem
- Cerebellum



Pathophysiology

Reduced blood flow



Oxygen deprivation



ATP depletion



Excitotoxicity



Cell death



Inflammation



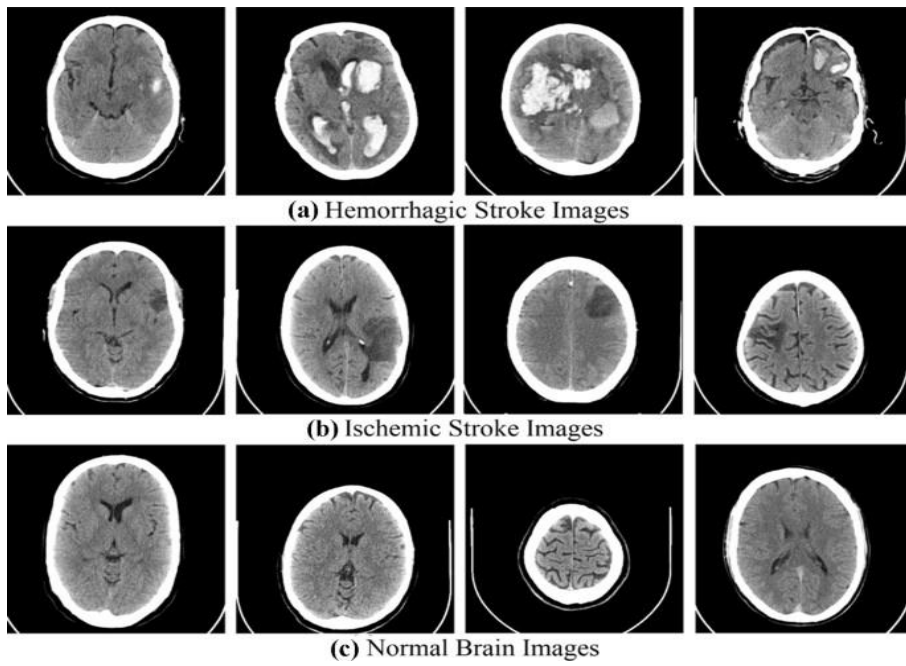
Brain infarction

Symptoms

- Facial drooping
- Arm weakness
- Speech difficulty
- Vision loss
- Hemiplegia
- Aphasia
- Ataxia

Diagnosis

- CT
- MRI
- CT angiography
- NIH Stroke Scale



Treatment

- Alteplase (tPA)
- Mechanical thrombectomy
- Antiplatelet therapy
- Blood pressure management
- Rehabilitation